SACHIN SAGAR

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SUMMARY

ADAS and Embedded Applications Engineer (B.Tech, IIT Madras) with 3 years of specialized R&D experience at Bajaj Auto. Expert in designing and deploying ADAS/EV control systems, sensor fusion (Radar/IMU), and real-time C/C++ applications for embedded platforms, like Arm and Renesas. Praised for AI automation of classical workflows and process improvement.

EDUCATION

Indian Institute of Technology Madras July 2019 – July 2023
Bachelor of Technology in Mechanical Engineering GPA: 8.92 Chennai, Tamil Nadu

EXPERIENCE

Bajaj Auto R&D June 2025 – Present
ADAS application developer *Pune, Maharashtra*

- Architected the inaugural ADAS perception and tracking stack on TI AWR1843 for L0 and L1 ADAS features across 2-3-4W passenger and commercial vehicles
- Engineered a real-time sensor fusion algorithm, synchronizing IMU, CAN bus odometry, and radar point clouds to achieve precise ego estimation and external object tracking for ADAS.
- Created a pipeline for converting point clouds to actionable ADAS decision variables for active safety features such as Active Emergency Braking and Adaptive Cruise Control comprising of clustering (DBScan) , association (JPDA) and tracking (IMM).
- Developed a Hardware-In-Loop (HIL) setup to simulate live data from vehicle and CARSIM on the MCU, cutting test time by 50%
- Optimized perception and tracking algorithms for constrained hardware, achieving zero heap allocation and <80kB stack usage, enhancing system reliability and efficiency.
- Leveraged AI coding assistants to develop internal software utilities, replacing commercial tools and realizing over 50LPA in annual licensing cost savings.

Bajaj Auto R&D Aug 2023 – June 2025
Controls Application Developer *Pune, Maharashtra*

- Developed a custom MATLAB Coordinate Transformation library with self-correction for IMU data normalization across SKUs, enabling advanced firmware features (e.g., roll-over detection) and reducing electronics BOM by 5%.
- Deployed a sensor-less mass and grade estimation algorithm using existing CAN bus data for shared mobility, preventing vehicle overloading and eliminating need for additional hardware/manpower.
- Developed a time-series application to model and correlate FET temperature against demand torque and motor rpm to derive a temperature estimate for an isolated MOSFET, eliminating the need for additional temperature sensor.
- Resolved 20+ critical defects in traction control and hill-assist logic through rigorous unit testing and system-level debugging

PROJECTS

Control of Camber Morphing Wing | Matlab, Python, C Oct 2022 – May 2023

- Spearheaded a research project in collaboration with DRDO to develop the control system for a composite camber morphing airfoil
- Architected a real-time control loop for a camber-morphing airfoil, modelled as a finite-state unsteady airfoil combining full-state feedback with a Kalman Filter to drive a Model Predictive Control (MPC) with a prediction horizon of 5 and control horizon of 2 steps
- Set up a Hardware in Loop (HIL) across MATLAB, Arduino, motor driver, and actuator for validating and improving the control logic

Road-Anomaly Detection & Geospatial Mapping | C/C++ Nov 2025 – Feb 2026

- Developed an Edge AI road-anomaly detection system using STM32 X-CUBE-AI to deploy an optimized Neural Network on an STM32 microcontroller; integrated 6-axis IMU (MPU6050) data for high-frequency vibration analysis and synchronized GPS (NEO-6M) coordinates to automate the geospatial mapping of road hazards like pot-holes and speed-bumps.

TECHNICAL SKILLS

Programming languages: C, C++,CUDA , Python, MATLAB, Simulink, Stateflow, Bash

Hardware and Com Protocols: TI mmWave Radar, ARM Cortex-M, Renesas RH850, R-Pi, ESP32, STM32, CAN, IPC, UART, SPI

Core Competencies: clustering, multi-object tracking, real-time object detection and classification, IMM-EKF, Sensor Fusion, PID, MPC, Kalman Filtering, Model Based Design

Engineering Tools: Code Composer Studio, TargetLink, Embedded Coder, Git, Linux, vSignalizer, RTOS, ROS2

ACHIEVEMENTS

- Awarded Bajaj Auto Innovision 2025 Process Pioneer Award for excellence in deploying complex workflows to production
- Awarded Honors in mechanical engineering and minors in Systems Engineering by HoD Mech. at IIT Madras 2023 convocation
- Awarded pre-placement offrer at Bajaj Auto for innovative breakthroughs during internship at the engine testing facility
- Winner of Engineering Design Competition (EV Category) at Formula Bharat 2022 representing Raftar Formula Racing (IIT Madras)